

Knowledge Graphs

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A Knowledge Graph is a structured representation of information where entities (people, places, objects, events, etc.) are connected through relationships. It stores knowledge in the form of a graph, making it easier for computers to understand and process information.

Knowledge Graphs are widely used in search engines, recommendation systems, chatbots, healthcare systems, and artificial intelligence applications.

Structure of Knowledge Graphs

1. **NODES** (vertices) represent entities (people, places, concepts, events).
2. **EDGES** represent relationships (directed links between entities).
3. **PROPERTIES/ATTRIBUTES** annotate nodes and edges with metadata.

Applications of Knowledge Graphs

1. **Search Engines:** Google Knowledge Graph improves search results by understanding relationships between entities.
2. **Recommendation Systems:** Netflix, Amazon recommend products, content using connected information, Chatbots, and Virtual Assistants.
3. **Healthcare:** Used to connect diseases, symptoms, medicines, and treatments.
4. **Social Networks:** Represent users and their interactions.

Tools

1. **NetworkX:** General-purpose graph library; great for KG construction/traversal
2. **RDFLib:** Python RDF/OWL library; supports SPARQL queries and serialization
3. **PyVis:** Interactive KG visualisation in HTML
4. **Owlready2:** OWL ontology loading and reasoning
5. **Neo4j (py2neo):** Property graph database backend
6. **GraphDB:** Enterprise triple store (SPARQL endpoint)